

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Transformations of Functions

Name the parent graph, then apply the transformation in the right direction.

TRANSFORMATIONS OF FUNCTIONS · P1 · 07

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 SHIFT: Translations

TRIGGER $f(x) + a, f(x - a)$

02 SCALE: Stretches

TRIGGER $af(x), f(ax)$

03 FLIP: Reflections

TRIGGER $-f(x), f(-x)$

04 ORDER: Combined Transformations

TRIGGER *more than one transformation*

BANK EVIDENCE Local pages: 33 problem, 50 answer, 18 notes. Website primary entries: 7.

01 SHIFT: Translations

TRIGGER $f(x) + a, f(x - a)$

BECOMES Move the whole graph without changing its shape.

FIRST LINE TO WRITE

$$y = f(x - a) + b$$

SIMPLEST STRATEGY

- 1 Start from $y = f(x)$.
- 2 Horizontal move comes from inside the bracket.
- 3 Inside sign works opposite.
- 4 Final vertical move is outside.
- 5 Translate every key point.

WORKED MODEL

$$y = f(x - 3) + 2$$

moves right 3 and up 2.

HARD VARIANTS

- 1 Move key points, endpoints, and asymptotes, not just the visible curve.
- 2 Inside bracket moves are opposite: $f(x - a)$ moves right a .
- 3 Update domain and range after the shift.

— BOTTOM LINE

Inside changes are horizontal and opposite.

02 SCALE: Stretches

TRIGGER $af(x), f(ax)$

BECOMES Multiply distances from an axis.

FIRST LINE TO WRITE

$$y = af(x), \quad y = f(ax)$$

SIMPLEST STRATEGY

- 1 Select vertical or horizontal.
- 2 Check whether the multiplier is outside or inside.
- 3 Apply reciprocal effect for inside.
- 4 Locate transformed key points.
- 5 End with the same shape.

WORKED MODEL

$$y = 2f(x)$$

doubles y-values; $y = f(2x)$ halves x-values.

HARD VARIANTS

- 1 Inside scale factors act reciprocally on x-coordinates.
- 2 Vertical stretch changes range; horizontal stretch changes domain/endpoints.
- 3 Asymptotes and intercepts must be stretched too.

— BOTTOM LINE

Inside stretch factors act reciprocally.

03 FLIP: Reflections

TRIGGER $-f(x), f(-x)$

BECOMES Reflect in the correct axis.

FIRST LINE TO WRITE

$$y = -f(x), \quad y = f(-x)$$

SIMPLEST STRATEGY

- 1 Find whether the minus is outside or inside.
- 2 Link outside to x-axis.
- 3 Inside to y-axis.
- 4 Plot reflected key points.

WORKED MODEL

$$y = -f(x)$$

reflects in the x-axis; $y = f(-x)$ reflects in the y-axis.

HARD VARIANTS

- 1 Outside minus reflects in the x-axis; inside minus reflects in the y-axis.
- 2 Reflect endpoints and asymptotes together with the graph.
- 3 Combined reflection and shift: transform key points in order.

— BOTTOM LINE

Outside controls y ; inside controls x .

04 ORDER: Combined Transformations

TRIGGER *more than one transformation*

BECOMES Apply inside changes to x-values and outside changes to y-values.

FIRST LINE TO WRITE

$$(x, y) \mapsto (\text{new } x, \text{new } y)$$

SIMPLEST STRATEGY

- 1 Organise inside and outside changes.
- 2 Rewrite key points.
- 3 Do horizontal changes carefully.
- 4 Execute vertical changes.
- 5 Redraw with labelled points.

WORKED MODEL

| For $y = 2f(x - 1) - 3$, point (a, b) becomes $(a + 1, 2b - 3)$.

HARD VARIANTS

- 1 For a point (a, b) , transform x using inside changes and y using outside changes.
- 2 Do not rely on drawing first; calculate key-point images first.
- 3 Check the final domain and range.

— BOTTOM LINE

Transform key points; the curve follows.

SAME IDEA Use ORDER: Combined Transformations.

QUESTION

| If $(2, 5)$ is on $y = f(x)$, find its image on $y = 3f(x - 4)$.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
$f(x) + a, f(x - a)$	$y = f(x - a) + b$
$af(x), f(ax)$	$y = af(x), \quad y = f(ax)$
$-f(x), f(-x)$	$y = -f(x), \quad y = f(-x)$
more than one transformation	$(x, y) \mapsto (\text{new } x, \text{new } y)$